

Weyl expansion of spherical wave

We wish to prove **Equation 17** of [Coupled electric and magnetic dipole formulation for planar arrays of particles: Resonances and bound states in the continuum for all-dielectric metasurfaces](https://journals.aps.org/prb/pdf/10.1103/PhysRevB.102.125411)

(<https://journals.aps.org/prb/pdf/10.1103/PhysRevB.102.125411>) and, equivalently, **Equation A4** of [Optical response features of Si-nanoparticle arrays](https://journals.aps.org/prb/pdf/10.1103/PhysRevB.82.045404)

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We do so by noting the following:

$$e^{ikr}/4\pi r = \frac{1}{8\pi^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{e^{i\mathbf{q}\cdot\mathbf{r}}}{|\mathbf{q}|^2 - k^2} d^3\mathbf{q},$$

which follows directly from the fact that both satisfy $(\nabla^2 + k^2)\psi(\mathbf{r}) = -\delta(\mathbf{r})$

Usually, in order to transform the RHS of this relation to resemble the form on the LHS, one computes the integral over the norm, $|\mathbf{q}|$, in spherical coordinates. In order to prove the Weyl expansion, we instead compute the integral over one wavevector, q_z :

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{e^{iq_z z}}{q_z^2 - (k^2 - q_y^2 - q_x^2)} dq_z &= \int_{-\infty}^{\infty} \frac{e^{iq_z z}}{\left(q_z - \sqrt{k^2 - q_y^2 - q_x^2}\right) \left(q_z + \sqrt{k^2 - q_y^2 - q_x^2}\right)} dq_z \\ &= \frac{1}{2\sqrt{k^2 - q_x^2 - q_y^2}} \int_{-\infty}^{\infty} e^{iq_z z} \left[\frac{1}{q_z - \sqrt{k^2 - q_x^2 - q_y^2}} - \frac{1}{q_z + \sqrt{k^2 - q_x^2 - q_y^2}} \right] dq_z \end{aligned}$$

If $k^2 < q_y^2 + q_x^2$, we have that $\sqrt{k^2 - q_x^2 - q_y^2}$ is purely imaginary. For $z > 0$, we close the contour in the upper half of the complex plane. For $z < 0$, we close the contour in the lower half of the complex plane, giving us:

$$\int_{-\infty}^{\infty} \frac{e^{iq_z z}}{\left(q_z - \sqrt{k^2 - q_y^2 - q_x^2}\right) \left(q_z + \sqrt{k^2 - q_y^2 - q_x^2}\right)} dq_z = \frac{i\pi e^{i|z|\sqrt{k^2 - q_x^2 - q_y^2}}}{\sqrt{k^2 - q_x^2 - q_y^2}}$$

If, on the other hand, $k^2 > q_x^2 + q_y^2$, we note that we must take $k \rightarrow k + i\delta$ in order to move the poles away from the real axis (this is also consistent with causality since we may not have a Green's function that is not analytic in the upper half of the complex plane). The integral we compute is thus:

$$\int_{-\infty}^{\infty} \frac{e^{iq_z z}}{\left(q_z - i\eta - \sqrt{k^2 - q_x^2 - q_y^2}\right) \left(q_z + i\eta + \sqrt{k^2 - q_x^2 - q_y^2}\right)} dq_z = \frac{i\pi e^{i|z|\sqrt{k^2 - q_x^2 - q_y^2}}}{\sqrt{k^2 - q_x^2 - q_y^2}},$$

which is the same answer derived before. Therefore, we have:

$$e^{ikr}/4\pi r = \frac{i}{8\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{e^{iq_x x + iq_y y + i|z|\sqrt{k^2 - q_x^2 - q_y^2}}}{\sqrt{k^2 - q_x^2 - q_y^2}} dq_x dq_y$$

Sanity check

In order to check that the above does indeed satisfy $(\nabla^2 + k^2)\psi(\mathbf{r}) = -\delta(\mathbf{r})$, we need to show that $\partial_z \psi(x, y, z = 0^+) - \partial_z \psi(x, y, z = 0^-) = -\delta^{(2)}(x, y)$, where $\delta^{(2)}$ is the two-dimensional Dirac delta function. We check that this works below:

$$\begin{aligned} & \partial_z \left[\frac{i}{8\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{e^{iq_x x + iq_y y + i|z|\sqrt{k^2 - q_x^2 - q_y^2}}}{\sqrt{k^2 - q_x^2 - q_y^2}} dq_x dq_y \right]_{z=0^+} - \\ & \partial_z \left[\frac{i}{8\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{e^{iq_x x + iq_y y + i|z|\sqrt{k^2 - q_x^2 - q_y^2}}}{\sqrt{k^2 - q_x^2 - q_y^2}} dq_x dq_y \right]_{z=0^-} = \\ & -\frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{iq_x x + iq_y y} dq_x dq_y = -\delta^{(2)}(x, y), \end{aligned}$$

where the last equality follows from the Fourier expansion of the Dirac delta function.